

Projected Increase in HIV Incidence in 11 States if Ryan White Ends: A Simulation Study

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Objectives. To estimate the increase in HIV infections in 11 US states if Ryan White services are interrupted or ended.

Methods. We applied a population-level model of HIV transmission to 11 states. We represented the proportion of people with HIV receiving Ryan White AIDS drug assistance, outpatient health services, or support services, and simulated a loss of suppression in each category if services permanently end or return after delays of 1.5 or 3.5 years.

Results. Cessation of Ryan White services in 2025 was projected to result in 69 695 additional infections from 2025 to 2030 (95% credible interval [CrI] = 18 943, 123 628), 68% more (95% CrI = 18%, 118%) than if Ryan White were continued. Temporary interruptions of 1.5 and 3.5 years resulted in 26 951 (95% CrI = 7341, 47 534) and 53 594 (95% CrI = 14 645, 94 860) additional infections, respectively. Excess infections varied across states, from a 45% increase in Texas to 126% in Missouri.

Conclusions. Projected increases in HIV infections because of disruptions of Ryan White services threaten the progress made in curtailing the US HIV epidemic, illustrating the critical role Ryan White plays in preventing HIV transmission. (*Am J Public Health.* 2026;116(5):732–735. <https://doi.org/10.2105/AJPH.2025.308409>)

The Ryan White HIV/AIDS Program provides comprehensive services for over half of the 1.2 million people with HIV (PWH) in the United States, often functioning as a “payer of last resort” for those who would otherwise be unable to access care.¹ The program provides assistance through several mechanisms: payment for antiretroviral therapy through the AIDS Drug Assistance Program, organizational funding for routine outpatient ambulatory health services, and nonmedical support including case management and transportation.² Consequently, Ryan

White clients have higher rates of viral suppression, rendering clients unlikely to transmit HIV.^{2–4}

The Ryan White Program is administered by the Health Resources and Services Administration and operates under annual budgetary approval from Congress.¹ Ryan White funds (\$2.5 billion in fiscal year 2025) are distributed through grants to local jurisdictions and HIV care organizations.¹ Disruptions to Ryan White services could substantially increase HIV transmission in the United States. We sought to estimate the increase in HIV infections in

11 states if Ryan White services were interrupted or discontinued.

METHODS

The Johns Hopkins Epidemiologic and Economic Model is a dynamic, compartmental model of HIV transmission that stratifies the adult population by age, race/ethnicity, sex, and HIV status⁵ (Figure A, available as a supplement to the online version of this article at <http://www.ajph.org>). Model calibration has been described previously⁵; we simulated state-level HIV epidemics

using a similar process (see online Appendix; Github Repository: https://github.com/tfojo1/jheem_analyses.git).

We simulated HIV epidemics from January 2025 to December 2030 in 11 states, in which 63% of people diagnosed with HIV in the United States reside: Alabama, California, Florida, Georgia, Illinois, Louisiana, Mississippi, Missouri, New York, Texas, and Wisconsin. These states were chosen via author consensus to represent varied geographic regions, Medicaid expansion status, and prioritization within the national Ending the HIV Epidemic initiative.⁶

A prior analysis examining the impact of Ryan White Program cessation at the city level describes our methodology for representing program services.⁷ We simulated the proportion of PWH receiving services in 3 mutually exclusive categories: (1) AIDS drug assistance (regardless of whether clients also received other services), (2) outpatient ambulatory health services at HIV care facilities (excluding those receiving drug assistance), and (3) other Ryan White support services (but not drug assistance or outpatient ambulatory health services; online Figure B).² Calibration targets for Ryan White service utilization included the number of clients accessing each service type and the proportion of clients who were virally suppressed in each state (Table B and Figure C).

Based on a survey of Ryan White clinic directors, administrators, and public health officials who estimated the proportion of clients who would lose viral suppression if services stopped, we assumed a mean 65% (interquartile range [IQR] = 40%–90%) loss of viral suppression among recipients of AIDS drug assistance, 49% (IQR = 25%–70%) among other outpatient health services recipients, and 37% (IQR = 10%–60%) among

Ryan White clients receiving other services⁷ (Table A). We ran 1000 simulations in each state, sampling the effects of ending services from the range of survey responses, conditional on Medicaid expansion status. Because survey respondents might overestimate the impact of disruptions, we performed a “conservative” secondary analysis in which we also incorporated data from 2 published studies on the association between Ryan White services and viral suppression.^{7–9}

We modeled 3 scenarios in which services stopped in July 2025 and viral suppression falls: (1) “cessation,” in which suppression never recovers; (2) “brief” (1.5-year) interruption, where suppression recovers from January to December 2027; and (3) “prolonged” (3.5-year) interruption, where suppression recovers from January to December 2029.

RESULTS

Assuming Ryan White services continue, we projected 103 101 new HIV infections (95% credible interval [CrI] = 97 773, 109 048) from 2025 to 2030 across all 11 states. If Ryan White services end permanently in 2025 (“cessation”), we projected 69 695 additional infections (95% CrI = 18 943, 123 628) from 2025 to 2030, an excess of 68% (95% CrI = 18%, 118%; Figure 1 and Figures D and E). In the “brief” (1.5-year) interruption scenario, we projected 26 951 (95% CrI = 7341, 47 534) additional infections, an excess of 26% (95% CrI = 7%, 46%). In the “prolonged” (3.5-year) interruption scenario, we projected 53 594 (95% CrI = 14 645, 94 860) additional infections, an excess of 52% (95% CrI = 14%, 90%). In the “conservative” analysis, additional infections were approximately half of those

from the primary analysis for all scenarios (Figures F and G). Infections increased most among adults aged younger than 35 years and men who have sex with men, without large differences by race (Table D).

Projected excess HIV infections varied across states (Figure 1), from 45% (95% CrI = 10%, 88%) in Texas to 126% (95% CrI = 25%, 246%) in Missouri (for individual states, see <https://jheem.org/ryan-white-state-level>). New infections would more than double in 4 states: Missouri, Alabama, Wisconsin, and Illinois. States with a greater proportion of PWH receiving Ryan White services in 2025 had more excess HIV infections (Figure H). In the 5 modeled Medicaid expansion states, fewer clients lost viral suppression, but because clients in those states had higher baseline suppression rates than in the 6 nonexpansion states, the relative increase in the number of unsuppressed PWH was similar, leading to similar increases in infections (Figure 1).

DISCUSSION

We used a validated HIV transmission model, informed by Centers for Disease Control and Prevention surveillance data and Ryan White Program reports, to project the impact of disruptions to the US Ryan White HIV/AIDS Program in 11 US states from 2025 to 2030. Permanently ending these services would lead to nearly 70 000 additional HIV infections in the next 5 years. Even temporary interruptions could result in a rise of over 26 000 additional infections. Impacts are expected to vary substantially at the state level, with the largest increases in new HIV infections in Missouri, Alabama, Wisconsin, and Illinois.

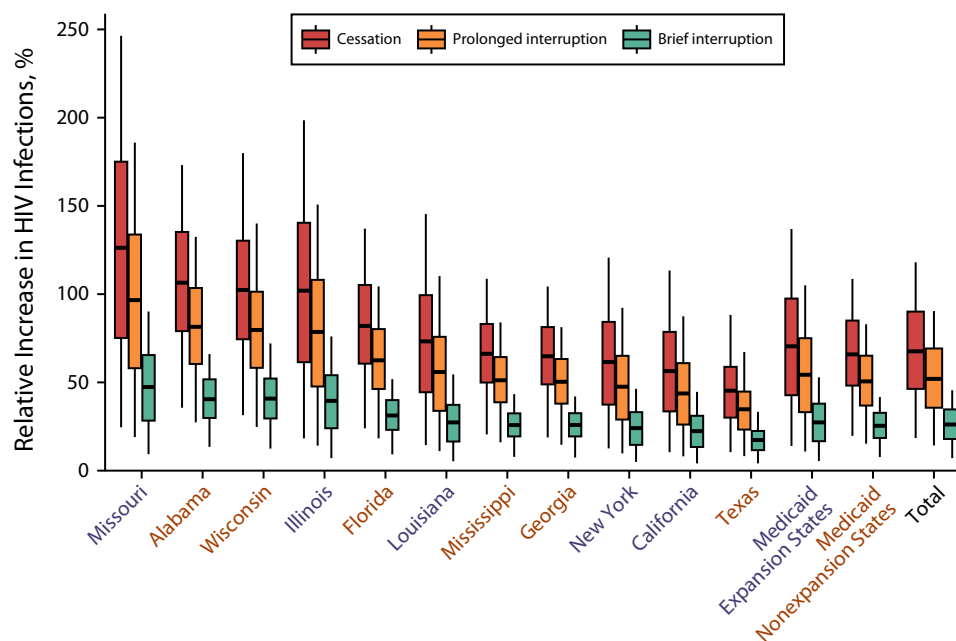


FIGURE 1— State-Level Relative Excess HIV Infections From 2025 to 2030 if Ryan White Programs Are Stopped or Interrupted

Note. Boxplots display the projected percentage increase in new HIV infections under 3 scenarios in which all Ryan White services stop in July 2025: “Cessation” (red)—viral suppression among Ryan White clients never recovers; “Prolonged [3.5-year] Interruption” (orange)—viral suppression among Ryan White clients recovers from January to December 2029; “Brief [1.5-year] Interruption” (green)—viral suppression among Ryan White clients recovers from January to December 2027. The value along the y-axis represents the relative increase in cases versus a scenario in which Ryan White services continue uninterrupted. The dark horizontal lines indicate the median projection across 1000 simulations, the boxes indicate interquartile ranges (IQR), and whiskers cover the 95% credible interval. Medicaid nonexpansion states are denoted by orange labels; expansion states are indicated by purple ones.

Our analysis is subject to limitations, especially as the individual-level effects of ending Ryan White services are uncertain. Without empirical evidence on these effects, we relied on estimates from those with expertise in Ryan White programming. We captured the uncertainty around these estimates by simulating effects across the range of survey responses. The wide credible intervals around our projections reflect the variation in these responses, which ranged from almost no effect of Ryan White cessation to nearly 100% losses in viral suppression. Our secondary, “conservative” analysis avoids potential bias in survey responses, but likely understates the causal effect of widespread cessation of Ryan White services. Second, we assume that cessation or interruption of programming will only

affect individuals accessing services. However, the implications of funding cuts to clinics (such as reducing staff or limiting services) will likely also affect the non-Ryan White clients who receive care at these locations. From this perspective, our estimates may underestimate the true impacts of reducing Ryan White funding. Finally, although we limited the scope of our analysis to estimate impacts on future transmission, interrupting Ryan White services would also affect both quality of life and mortality for those currently living with HIV.

The modeling approach used here has several strengths. Our Bayesian calibration process is a rigorous approach to handling unknown variables and missing data in HIV epidemiology. Furthermore, by calibrating to stratified epidemiological data at the state level,

we can compare across settings and capture local differences in HIV epidemics.

In summary, projected increases in new HIV infections threaten progress made in curtailing the US HIV epidemic, should Ryan White services be interrupted or discontinued. These findings illustrate the critical role that Ryan White services play in preventing the transmission of HIV and the importance of maintaining these programs at the state level. [AJPH](#)

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M. Schnure made substantial contributions to study conceptualization and design, conducted the formal analyses, and led interpretation of results; she drafted the initial manuscript and integrated critical revisions from all coauthors. R. Forster made substantial contributions to study conceptualization and design, analytic approach, and interpretation of findings as a core member of the analytic team; he provided critical revisions that shaped the intellectual content of the manuscript. J. L. Jones contributed to study question development and contributed clinical and public health expertise that informed interpretation of findings and their implications for practice; she critically revised the manuscript. C. R. Lesko contributed to refinement of the study question, methodological considerations, and epidemiological interpretation of results; she critically revised the manuscript. D. S. Batey contributed substantial expertise related to HIV service delivery and policy implementation, informing interpretation of findings; he critically revised the manuscript. I. Butler contributed clinical and implementation expertise relevant to HIV care delivery, informing

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CONFLICTS OF INTEREST

The authors have no conflicts of interest to declare.

HUMAN PARTICIPANT PROTECTION

This study consisted solely of mathematical modeling using publicly available data and did not involve interaction with human participants; therefore, institutional review board approval was not required.

REFERENCES

1. KFF. The Ryan White HIV/AIDS Program: the basics. 2022. Available at: <https://www.kff.org/>

[hiv-aids/fact-sheet/the-ryan-white-hiv-aids-program-the-basics](#). Accessed May 6, 2025.

2. Bradley H, Viall AH, Wortley PM, Dempsey A, Hauck H, Skarbinski J. Ryan White HIV/AIDS program assistance and HIV treatment outcomes. *Clin Infect Dis*. 2016;62(1):90–98. <https://doi.org/10.1093/cid/civ708>
3. Cohen MS, Chen YQ, McCauley M, et al. Antiretroviral therapy for the prevention of HIV-1 transmission. *N Engl J Med*. 2016;375(9):830–839. <https://doi.org/10.1056/NEJMoa1600693>
4. Rodger AJ, Cambiano V, Bruun T, et al. Risk of HIV transmission through condomless sex in serodifferent gay couples with the HIV-positive partner taking suppressive antiretroviral therapy (PARTNER): final results of a multicentre, prospective, observational study. *Lancet*. 2019;393(10189):2428–2438. [https://doi.org/10.1016/S0140-6736\(19\)30418-0](https://doi.org/10.1016/S0140-6736(19)30418-0)
5. Fojo AT, Schnure M, Kasaie P, Dowdy DW, Shah M. What will it take to end HIV in the United States? A comprehensive, local-level modeling study. *Ann Intern Med*. 2021;174(11):1542–1553. <https://doi.org/10.7326/M21-1501>
6. HIV.gov. Ending the HIV epidemic: plan for America. 2023. Available at: <https://www.hiv.gov/federal-response/ending-the-hiv-epidemic/overview>. Accessed November 10, 2020.
7. Forster R, Schnure M, Jones J, et al. The potential impact of ending the Ryan White HIV/AIDS Program on HIV Incidence: a simulation study in 31 US cities. *Ann Intern Med*. 2025;178(11):1580–1590. <https://doi.org/10.7326/ANNALS-25-01737>
8. Erly SJ, Khosropour CM, Hajat A, Sharma M, Reuer JR, Dombrowski JC. AIDS Drug Assistance Program disenrollment is associated with loss of viral suppression beyond differences in homelessness, mental health, and substance use disorders: an evaluation in Washington State 2017–2019. *PLoS One*. 2023;18(5):e0285326. <https://doi.org/10.1371/journal.pone.0285326>
9. Diepstra KL, Rhodes AG, Bono RS, Patel S, Yerkes LE, Kimmel AD. Comprehensive Ryan White assistance and human immunodeficiency virus clinical outcomes: retention in care and viral suppression in a Medicaid nonexpansion state. *Clin Infect Dis*. 2017;65(4):619–625. <https://doi.org/10.1093/cid/cix380>